TATA POWER



EV Charging Infrastructure

Performance, Profitability & Expansion Strategy — Analytical Report

A Data-Driven View Across Operations, Customers & Financial ROI
to Guide Investment Decisions Across Districts

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Executive Summary

Four analytical lenses on the EV charging network — from usage to unit economics

01

Business & Usage Overview

How demand is evolving, whether stations keep pace, and daily charging behaviour.

02

Station Utilization & Efficiency

Over- and under-utilized stations, plug efficiency, and district-level load balance.

03

Customer & Demand Behavior

Who drives usage and revenue, how income tier and EV model shape charging habits.

04

Financial Performance & Expansion (ROI)

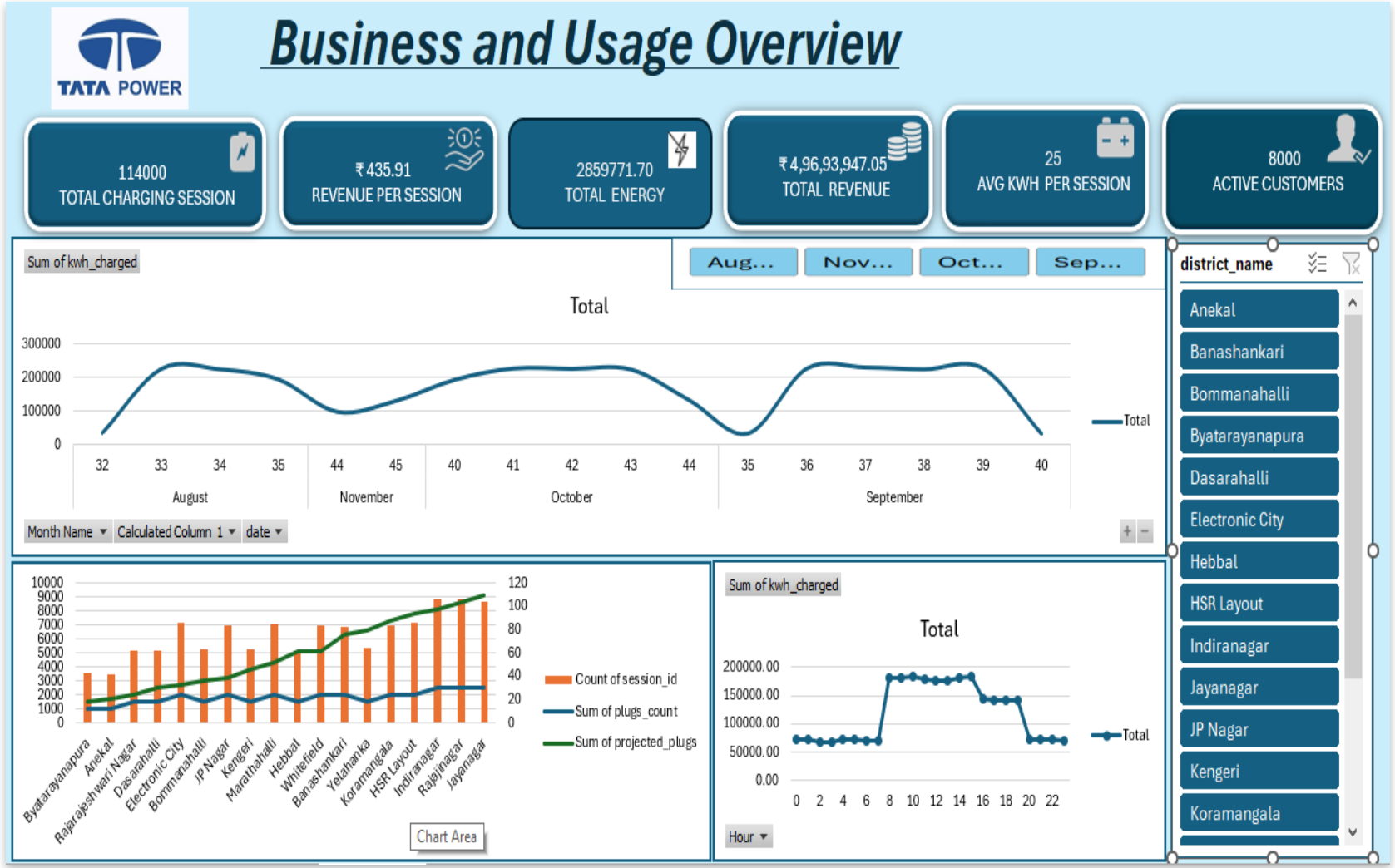
CAPEX, payback, EBITDA and which districts to prioritize for new stations.

Headline Numbers

114,000	28.6 Lakh	₹4.97 Cr	8,000	41%	4.43 Yrs
Total Charging Sessions	kWh Charged (Total Energy)	Total Revenue	Active Customers	5-Yr ROI	Payback (with Subsidy)

Page 1 · Business & Usage Overview

Dashboard snapshot — demand trend, session growth and hourly usage pattern



1,14,000
Total Charging Sessions

₹435.91
Revenue per Session

28,59,771.70
Total Energy (kWh Charged)

₹4,96,93,947.05
Total Revenue

25
Avg. kWh per Session

8,000
Active Customers

Page 1 · KPI Insights

Business and Usage Overview — what the numbers mean

Total Charging Sessions 1,14,000

Confirms strong, sustained adoption across the network — averages roughly 9,500 sessions per month over the reporting window.

Revenue per Session ₹435.91

A healthy per-visit ticket size for a mixed AC/DC fleet, suggesting pricing is capturing fair value per charge rather than under-monetizing sessions.

Total Energy (kWh Charged) 28,59,771.70

At a 25 kWh/session average, this points to mostly partial top-up charging rather than full-battery sessions across the base.

Total Revenue ₹4,96,93,947.05

Nearly ₹5 Cr from charging alone — the revenue base that CAPEX and EBITDA figures on Page 4 should be benchmarked against.

Avg. kWh per Session 25

Typical of opportunistic daytime top-ups; supports designing tariffs and slot durations around shorter sessions rather than long full charges.

Active Customers 8,000

With 1.14L sessions this implies ~14 sessions/customer — matching Page 3's KPI exactly, and confirming a loyal, repeat-usage base rather than one-off users.

Page 1 · Insights & Answers

Business and Usage Overview — pivot-based findings

1 How is charging demand evolving over time?

The weekly kWh-charged trend (Aug–Nov) is cyclical rather than linear: it climbs to a peak around week 33 (Aug), dips through weeks 34–35, recovers to a second, slightly higher peak across weeks 36–39 (Sept), and tapers again into October–November. Demand is healthy and recurring but not yet on a sustained upward trajectory — it moves in seasonal waves.

2 Are stations able to handle growing demand?

Plotting session count, plug count and projected plugs by district shows session volumes and projected-plug requirements rising together across districts (Byatarayanapura through Jayanagar/Rajajinagar), while actual deployed plugs grow more slowly. In the highest-session districts the gap between demand and installed capacity widens — a sign that infrastructure additions are lagging real usage growth.

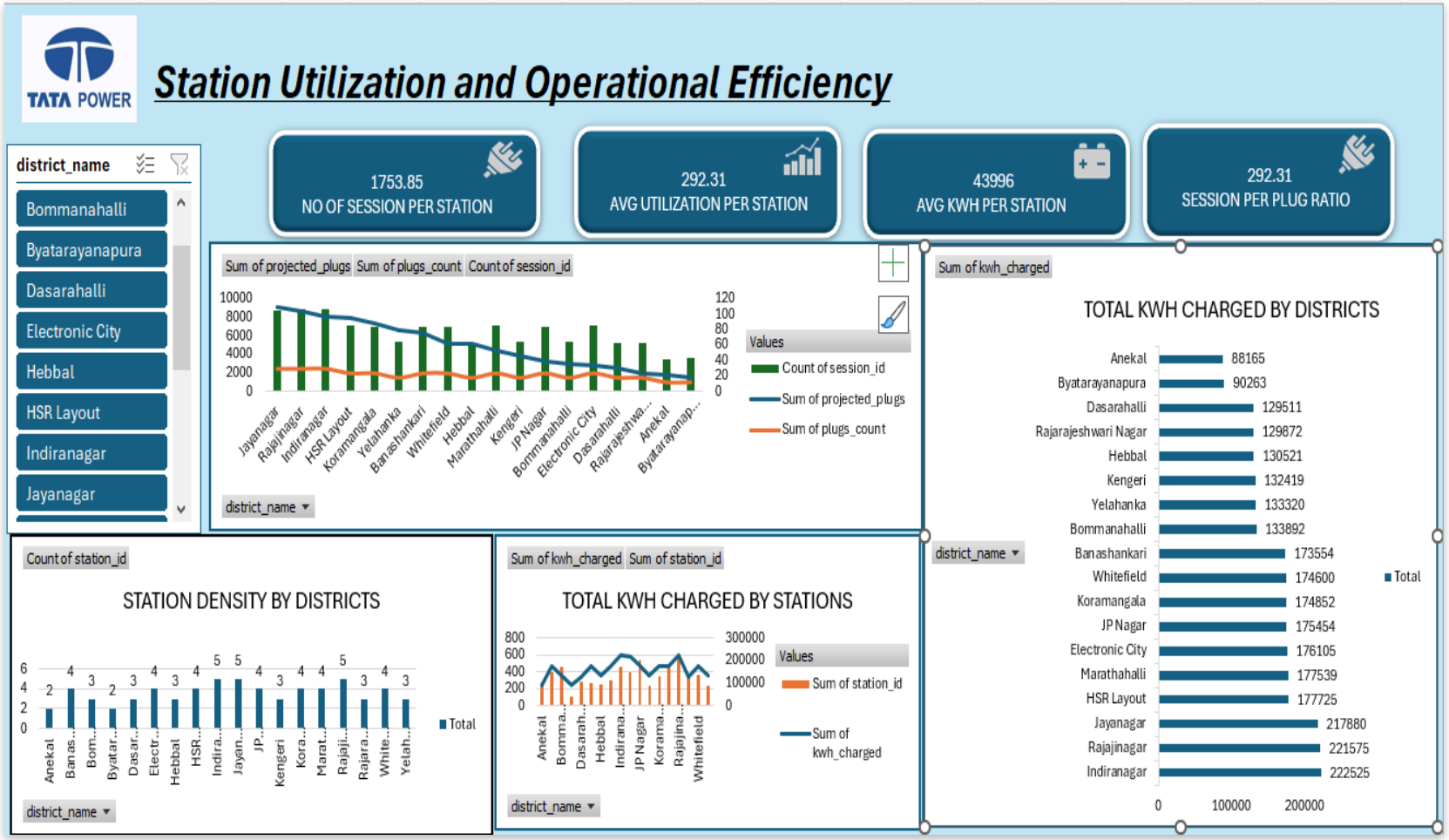
3 What is the charging behaviour pattern by time of day?

Hourly kWh-charged data shows near-zero charging from midnight to 7 AM, a sharp jump at 8 AM, and sustained high, stable usage through roughly 8 PM before tapering off overnight again. Charging is overwhelmingly a daytime/working-hours activity — pricing, staffing and load-balancing should be optimized for the 8 AM–8 PM window rather than spread evenly across 24 hours.

Takeaway: Demand is cyclical and concentrated in daytime hours — capacity planning should track district-level projected-plug gaps, not just aggregate totals.

Page 2 · Station Utilization & Operational Efficiency

Dashboard snapshot — session load, plug usage and district density



1,753.85
Sessions per Station

292.31
Avg. Utilization per Station

43,996
Avg. kWh per Station

292.31
Sessions per Plug Ratio

Page 2 · KPI Insights

Station Utilization and Operational Efficiency — what the numbers mean

Sessions per Station 1,753.85

High throughput per station in aggregate — the network overall is well-used, even though the district-level view shows this load is unevenly distributed.

Avg. Utilization per Station 292.31

A substantial number of sessions per plug — at this level many plugs are close to practical capacity, reinforcing the case for adding plugs in the busiest districts.

Avg. kWh per Station 43,996

Confirms each station is moving meaningful energy volume, supporting the station-level revenue and payback assumptions used in the Page 4 ROI model.

Sessions per Plug Ratio 292.31

Identical to avg. utilization — plugs are working hard. Combined with the projected-plug gap on Page 1, this is the clearest signal that plug supply, not station count, is the binding constraint.

Page 2 · Insights & Answers

Station Utilization and Operational Efficiency — pivot-based findings

1 Which stations are overutilized?

By total kWh charged, Indiranagar (2,22,525), Rajajinagar (2,21,575) and Jayanagar (2,17,880) top every district — these carry the heaviest load and are the strongest candidates for added plugs/capacity expansion.

2 Which stations are underutilized?

Anekal (88,165 kWh), Byatarayanapura (90,263) and Dasarahalli (1,29,511) sit at the bottom of the district ranking — these need demand-generation, pricing incentives or route/marketing support rather than new hardware.

3 Are available plugs efficiently used?

Session counts and projected-plug requirements track closely across districts, but actual plug counts are comparatively flatter — plug supply isn't scaling with real usage in the busiest districts, pointing to uneven, inefficient utilization of the existing plug base.

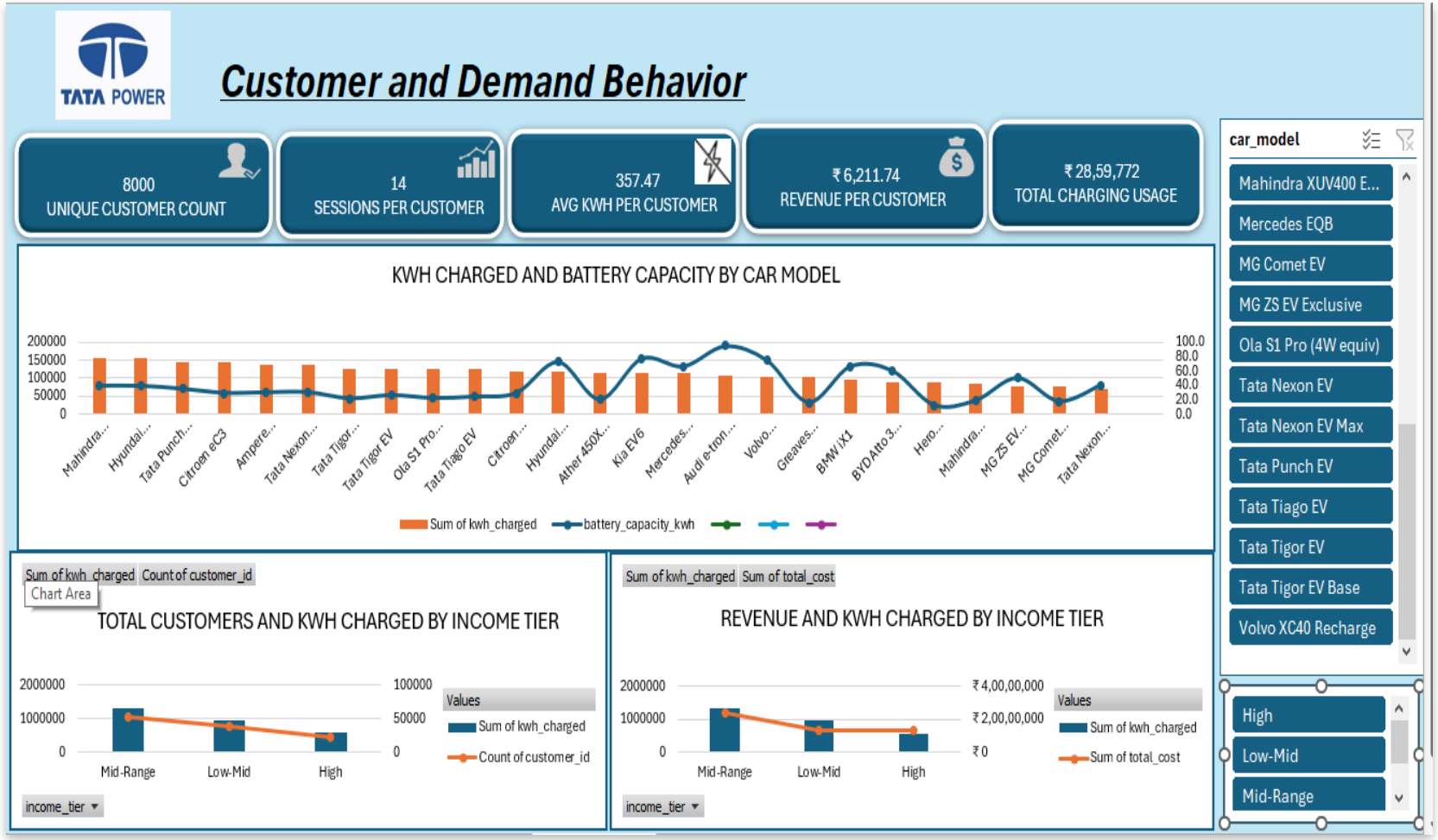
4 What is the load distribution across districts and stations?

Station density ranges from 2 to 6 per district (Yelahanka, Whitefield and JP Nagar lead), yet kWh usage doesn't scale directly with station count — e.g. Indiranagar posts the highest total kWh despite only moderate station density — signalling an uneven load per station across the network.

Takeaway: Expansion capital should follow the overutilized cluster (Indiranagar, Rajajinagar, Jayanagar) while underutilized districts (Anekal, Byatarayanapura) need demand-side interventions before more hardware is added.

Page 3 · Customer & Demand Behavior

Dashboard snapshot — income tier, car model and battery-capacity usage



8,000
Unique Customer Count

14
Sessions per Customer

357.47
Avg. kWh per Customer

₹6,211.74
Revenue per Customer

28,59,772
Total Charging Usage (kWh)

Page 3 · KPI Insights

Customer and Demand Behavior — what the numbers mean

Unique Customer Count 8,000

Matches Page 1's active-customer figure exactly — a stable, well-defined customer base that can be segmented and targeted with confidence.

Sessions per Customer 14

Indicates healthy repeat engagement rather than one-time trial usage — a solid foundation for loyalty programs or subscription-style pricing.

Avg. kWh per Customer 357.47

Roughly consistent with 14 sessions × 25 kWh/session from Page 1 — the customer- and session-level averages cross-check cleanly.

Revenue per Customer ₹6,211.74

Implies about ₹444 of revenue per session per customer, closely matching the network-wide revenue-per-session KPI (₹435.91) — consistent unit economics.

Total Charging Usage 28,59,772 kWh

Same order of magnitude as Page 1's total energy KPI — a useful cross-check that customer-level and network-level rollups reconcile.

Page 3 · Insights & Answers

Customer and Demand Behavior — pivot-based findings

1 **What type of customers contribute most to usage and revenue?**

Mid-Range income tier customers post the highest kWh charged and the largest customer count, followed by Low-Mid; High-income customers are the smallest group by volume and revenue despite a presumably higher per-visit spend capacity.

2 **How does income tier influence charging behaviour?**

Both total kWh charged and total revenue fall steadily from Mid-Range to Low-Mid to High tier — Mid-Range drives the bulk of volume and revenue, while the High tier, though fewer in number, likely represents higher-value, lower-frequency usage.

3 **What are the usage patterns across EV models and battery capacities?**

kWh charged is fairly even across car models (Mahindra, Hyundai, Tata Punch/Nexon/Tigor EV, Citroen eC3, Kia EV6, BYD Atto, Volvo XC40 and others), while battery capacity varies more, peaking for models like the Ather 450X and Kia EV6 — demand is broad-based, not concentrated in one segment.

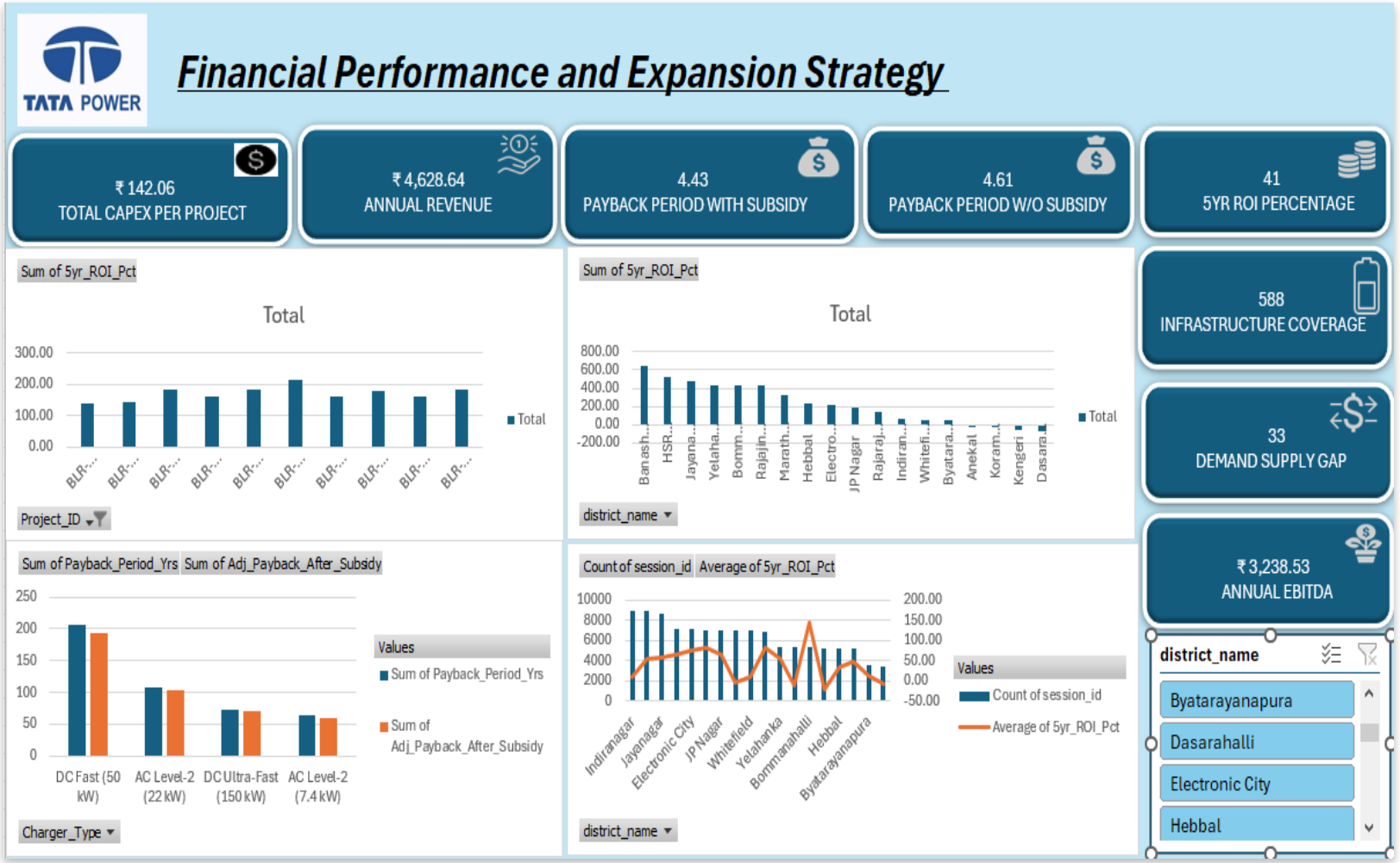
4 **Are there high-potential but low-engagement segments?**

The High income tier stands out: it has the fewest customers and the lowest total kWh/revenue contribution despite the greatest likely purchasing power for premium or fast-charging services — a strong candidate for targeted premium plans and engagement campaigns.

Takeaway: Mid-Range tier is the volume engine today; the High-income tier is under-tapped and worth a dedicated premium acquisition push to lift revenue per customer.

Page 4 · Financial Performance & Expansion Strategy (ROI)

Dashboard snapshot — CAPEX, payback, ROI and district-level returns



₹142.06 L
Total CAPEX per Project

₹4,628.64 L
Annual Revenue

4.43 Yrs
Payback Period (with Subsidy)

4.61 Yrs
Payback Period (w/o Subsidy)

41%
5-Yr ROI Percentage

588
Infrastructure Coverage

33
Demand-Supply Gap

₹3,238.53 L
Annual EBITDA

Page 4 · KPI Insights

Financial Performance and Expansion Strategy — what the numbers mean

Total CAPEX per Project ₹142.06 L

The benchmark capital outlay per new site — the base figure every payback and ROI calculation on this page is built from.

Annual Revenue ₹4,628.64 L

Set against EBITDA of ₹3,238.53L, this implies a strong ~70% EBITDA margin — confirming the underlying model is highly profitable once a station is running.

Payback Period (with Subsidy) 4.43 Yrs

Comfortably inside a typical 5-year investment horizon for infrastructure — supports continued rollout at current CAPEX levels.

Payback Period (w/o Subsidy) 4.61 Yrs

Only ~0.18 years (about 2 months) longer than the subsidized case — projects stay viable even without subsidy support, just slower to recover.

5-Yr ROI Percentage 41%

Works out to roughly 8% annualized return — a solid, bankable profile that should keep this asset class attractive to internal capital allocators.

Infrastructure Coverage 588

The current installed base to weigh against projected EV growth when deciding where next to add stations or plugs.

Demand-Supply Gap 33

A modest ~5.6% shortfall relative to the 588 coverage figure — manageable today, but one that should be closed before demand outpaces it further.

Annual EBITDA ₹3,238.53 L

Strong absolute profitability at the portfolio level, and the numerator behind the healthy ~70% margin and 41% five-year ROI reported alongside it.

Page 4 · Insights & Answers

Financial Performance and Expansion Strategy — pivot-based findings

1 Which districts/projects deliver the highest ROI?

Banashankari leads 5-yr ROI% by a wide margin, followed by HSR Layout, Jayanagar, Yelahanka, Bommanahalli and Rajajinagar. Dasarahalli, Koramangala and Kengeri sit at the bottom — expansion capital should be weighted toward the top-ROI cluster.

2 What is the payback period for different charging setups?

DC Fast (50kW) chargers show the longest payback, followed by AC Level-2 (22kW), then DC Ultra-Fast (150kW) and AC Level-2 (7.4kW) with the shortest. The subsidy-adjusted payback is consistently lower than the base payback for every charger type, confirming subsidies materially shorten recovery time.

3 How do CAPEX and related costs impact profitability?

Against a Total CAPEX of ₹142.06L per project, Annual EBITDA of ₹3,238.53L and Annual Revenue of ₹4,628.64L, the portfolio delivers a 41% five-year ROI with payback inside 5 years (4.43 yrs with subsidy, 4.61 yrs without) — CAPEX intensity is being recovered quickly enough to support continued rollout.

4 Which locations should be prioritized for new deployment?

Cross-referencing the top-ROI districts (Banashankari, HSR Layout, Jayanagar) with high session-load districts from Page 2 (Indiranagar, Rajajinagar) gives a combined demand + returns view — these districts should be first in line for new station capital.

5 What is the gap between existing infrastructure and projected EV demand?

Infrastructure coverage stands at 588 against a demand-supply gap of 33 (projected EVs vs. projected plugs) — a real but manageable shortfall concentrated in the highest-session districts, where projected-plug lines outpace actual plug counts on the Page 1/2 trend charts.

Priority Tiers for Expansion: Tier 1 (invest first) – Banashankari, HSR Layout, Jayanagar, Indiranagar, Rajajinagar | Tier 2 (monitor & grow) – Yelahanka, Bommanahalli, Whitefield | Tier 3 (optimize before expanding) – Dasarahalli, Koramangala, Kengeri, Anekal.

Strategic Recommendations

What the data means for investment, pricing and operations

1

Expand in the Tier-1 cluster

Prioritize CAPEX in Banashankari, HSR Layout, Jayanagar, Indiranagar and Rajajinagar — highest ROI combined with highest existing load.

2

Right-size plugs to demand, not just stations

Close the projected-plug vs. actual-plug gap in high-session districts before adding brand-new sites.

3

Optimize, don't expand, low-utilization districts

Anekal, Byatarayanapura and Dasarahalli need pricing/marketing interventions to lift utilization before further build-out.

4

Design pricing and staffing around the 8 AM–8 PM peak

Charging is a daytime activity — dynamic pricing and maintenance windows should reflect this, not a flat 24-hour assumption.

5

Build a premium play for the High income tier

Smallest, most under-tapped customer segment by volume — a strong target for premium/fast-charging bundles to lift revenue per customer.

6

Keep leaning on subsidy-backed rollouts

Subsidy consistently cuts payback across every charger type — continue structuring new sites to qualify wherever possible.